PULLBACK ALONG A COMPOSITION IS THE COMPOSITION OF PULLBACKS BY ADJOINT UNIQUENESS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Left adjoints to a given functor are unique up to unique isomorphism. invoked at proof lines 2435-2438. Identifies the two left adjoints once the common right adjoint is computed.
- Pushforward along a composition equals the composition of pushforwards (lemma-pushforward-composition). invoked at proof lines 2436-2438. Gives the common right adjoint (g composition f)_* = g_* composition f_* that both functors are left adjoint to.

Source reference. Original lemma 008E: sources/stacks-sheaves/pinned-original-source.tex, source lines 2255–2255. Cited at proof lines 2436-2438 to supply the common right adjoint; load-bearing inside the counted span.

Lemma 0.2. *Let $f : X \to Y$ and $g : Y \to Z$ be continuous maps of topological spaces. The functors $(g \circ f)^{-1}$ and $f^{-1} \circ g^{-1}$ are canonically isomorphic. Similarly $(g \circ f)_p \cong f_p \circ g_p$ on presheaves.*

Proof. To see this use that adjoint functors are unique up to unique isomorphism, and Lemma 008E. □

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